

# NATURAL GAS TRANSMISSION MAINLINE - 3LPE EXTERNAL COATING QUALITY PLAN (ITP)

**Pipe & Coating Description:** 42" (1067 mm) x 15.90 mm, Grade X70M PSL 2 SAWH/LSAW with 3LPE External Coating (Class v Reinforced) | **Governing Codes:** ISO 21809-1 / DIN 30670 / API Spec 5L 47th Ed. / BOTAN Specification

| No | Inspection / Test Stage                                | Frequency & Sampling                          | Location & Specimen                    | Test Standard & Procedure             | Specified Acceptance Criteria & Tolerances                                                                                                                  | Governing Clause        | Hold / Witness |
|----|--------------------------------------------------------|-----------------------------------------------|----------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|
| 1  | <b>Raw Material Verification (HDPE, Adhesive, FBE)</b> | Every batch/delivery lot of coating materials | Manufacturer raw batch samples         | ISO 21809-1 Table 1 & DIN 30670       | FBE: Gel time, moisture <= 0.5%, DSC Tg; Adhesive: MFR, density; PE: Density >= 0.940 g/cm3, MFR 0.2-0.7 g/10min, carbon black 2.0-2.5%, EN 10204 3.1 certs | ISO 21809-1 Cl. 6       | H/R            |
| 2  | <b>Bare Pipe Receipt &amp; Pre-Washing</b>             | 100% bare pipe joints before entry            | External pipe surface                  | SSPC-SP 1 / ISO 8504                  | No oil, grease, salt, lacquers, slivers or surface gouges; high-pressure potable water wash to remove contaminants                                          | ISO 21809-1 Cl. 8.1     | W              |
| 3  | <b>Pipe Pre-Heating Prior to Blasting</b>              | Continuous monitoring all pipes               | Pipe exterior surface                  | Optical Pyrometer                     | Steel temperature 40 °C - 60 °C and minimum 3 °C above ambient dew point to ensure completely dry surface                                                   | DIN 30670 Cl. 4.2       | W              |
| 4  | <b>Abrasive Blast Cleaning</b>                         | 100% of all pipes (continuous run)            | Full external steel surface            | ISO 8501-1 / SSPC-SP 10               | Cleanliness: Sa 2.5 (Near-White Metal). Steel grit/shot mix; free of mill scale, rust, and surface defects                                                  | ISO 21809-1 Cl. 8.2     | H/W            |
| 5  | <b>Surface Anchor Profile Measurement</b>              | Twice per 8-hour shift (3 pipes each)         | Replica Tape (Press-O-Film)            | ISO 8503-2 / ASTM D4417 Method C      | Surface roughness profile: Rz 60 µm - 100 µm with sharp angular profile for optimal FBE mechanical interlock                                                | ISO 21809-1 Table 2     | W              |
| 6  | <b>Residual Dust Level Inspection</b>                  | At start of shift and every 2 hours           | Dust tape on blasted surface           | ISO 8502-3                            | Dust rating: Maximum Class 1 or Class 2; blown clean with dry oil-free compressed air                                                                       | ISO 8502-3              | W              |
| 7  | <b>Water-Soluble Salt Contamination Test</b>           | 3 tests per shift (at pipe ends & middle)     | Bresle Patch & conductivity meter      | ISO 8502-6 / ISO 8502-9               | Residual chloride/salt level: <= 20 mg/m² (2.0 µg/cm²). If exceeded, re-wash with DI water and re-blast                                                     | ISO 21809-1 Cl. 8.2.3   | H/W            |
| 8  | <b>Chemical Surface Pre-Treatment</b>                  | Continuous monitoring of bath & pipe          | Acid wash or chromate spray            | Supplier Spec / Deionised rinse       | Phosphoric acid or chromate pre-treatment followed by DI water rinse; surface conductivity < 50 µS/cm                                                       | ISO 21809-1 Cl. 8.2.4   | W              |
| 9  | <b>Induction Heating for FBE Coating</b>               | Continuous pyrometer & infrared logger        | Full pipe circumference                | Multi-point optical pyrometer         | Steel temperature 190 °C - 220 °C (maximum allowable 230 °C). Overheating > 240 °C causes rejection                                                         | ISO 21809-1 Cl. 8.3     | H/W            |
| 10 | <b>Fusion Bonded Epoxy (FBE) Application</b>           | Continuous electrostatic powder spray         | First layer on pre-heated steel        | Electrostatic spray guns              | Dry film thickness (DFT): 150 µm - 250 µm (target 200 µm). Uniform cloud, partial cure verification                                                         | ISO 21809-1 Table 2     | W              |
| 11 | <b>Grafted Copolymer Adhesive Extrusion</b>            | Continuous side-wrap extrusion                | Second layer over partially cured FBE  | Flat die / crosshead extruder         | Adhesive thickness: 150 µm - 250 µm (target 200 µm). Full fusion with FBE before epoxy full cure                                                            | ISO 21809-1 Table 2     | W              |
| 12 | <b>Polyethylene (HDPE) Topcoat Application</b>         | Continuous circumferential wrapping           | Outer third protective layer           | Side extrusion + silicone roller      | Total 3LPE thickness: Minimum 3.0 mm (Reinforced Class v / Class C); weld seam min 85% of nominal body                                                      | DIN 30670 / ISO 21809-1 | H/W            |
| 13 | <b>Water Quench Cooling &amp; Temperature</b>          | 100% coated pipes through quench trough       | Full pipe length                       | Infrared Thermometer                  | Controlled water cooling; pipe surface temperature reduced to < 60 °C before conveyor handling and stenciling                                               | DIN 30670 Cl. 4.4       | W              |
| 14 | <b>100% Online Holiday Detection</b>                   | Every coated pipe (100% full surface)         | Full 360° coating surface              | NACE SP0188 / ISO 21809-1             | High-voltage spark testing: 25 kV (or 5 kV/mm). ZERO holidays / pinholes permitted per pipe length                                                          | DIN 30670 Cl. 5.3       | H/W            |
| 15 | <b>Total Coating Thickness (DFT) Scanning</b>          | 100% of all pipes (4 points per length)       | Clock positions 12, 3, 6, 9 o clock    | Magnetic/Eddy-current thickness gauge | Minimum total thickness: >= 3.0 mm on body, >= 2.5 mm on weld seam. No under-thickness permitted                                                            | DIN 30670 Table 3       | W              |
| 16 | <b>Peel Adhesion (Peel Strength) Testing</b>           | 1 test per 50 pipes (or 1 per 4 hours)        | Test ring from cutback or pipe end     | ISO 21809-1 Annex C / DIN 30670       | At 23 °C: Minimum 35 N/cm; At 50 °C: Minimum 15 N/cm. Failure mode: Cohesive within adhesive/PE (no steel disbond)                                          | ISO 21809-1 Annex C     | H/W            |
| 17 | <b>Cathodic Disbondment Test</b>                       | 1 test per lot / qualification batch          | Laboratory coupon from production pipe | ISO 21809-1 Annex H / ASTM G8         | 28 days at 23 °C: Disbondment radius <= 7.0 mm; 48 hours at 65 °C: Disbondment radius <= 7.0 mm                                                             | ISO 21809-1 Table 4     | H/W            |
| 18 | <b>Impact Resistance &amp; Indentation Hardness</b>    | 1 test per lot / shift                        | Production pipe sample                 | ISO 21809-1 Annex E & F               | Impact resistance: >= 7.0 J/mm (zero holiday at 25 kV after impact); Indentation depth: <= 0.2 mm at 23 °C                                                  | DIN 30670 Cl. 5.5       | W              |

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|----|--------------------------------------|------------------------------------|---------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------|----------------|
| 19 | Elongation at Break (PE Layer)       | 1 test per raw material lot        | Dumbbell specimen from outer PE | ISO 527-2 / DIN 30670           | Tensile elongation at break: Minimum >= 400% at 23 °C (demonstrating cold-bending ductile flexibility)                 | DIN 30670 Table 2   | W              |
| 20 | End Cutback & Bevel Chamfer Geometry | 100% both ends of each pipe        | Pipe ends (internal & external) | Vernier Caliper / Depth Gauge   | Cutback length: 150 mm +/- 20 mm; Chamfer angle: <= 30°; bare steel free of coating residue, bevel protector installed | ISO 21809-1 Cl. 8.4 | W              |
| 21 | Coating Repair Inspection            | All repaired areas (if applicable) | Repaired defect patches         | ISO 21809-1 Cl. 8.5 / DIN 30670 | Max defect size <= 100 cm2, max 2 repairs per pipe. Melt stick + PE patch; 100% holiday re-tested at 25 kV             | DIN 30670 Cl. 5.6   | W              |
| 22 | Final Visual, Stenciling & Release   | 100% of all finished coated pipes  | Pipe body and internal bore     | Visual / Project Stencil Spec   | 100% visual inspection; UV-resistant white paint stencil; heavy-duty end caps fitted; EN 10204 3.1 3LPE cert           | ISO 21809-1 Cl. 11  | H/W            |